

Performance Testing

Date	17 JULY 2026
Team ID	SWTID-2026-4144
Project Name	RISING WATERS
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	[e.g., JMeter, Locust, k6, Postman]
Type of Testing	[e.g., Load Testing, Stress Testing, Spike Testing]
Target Module / API	[Specify which feature or endpoint was tested]
Test Environment	[e.g., Local / Cloud / Hosted Server]
Test Date	1-07-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Normal Load: Simulate regular users checking flood status	100	300	Response time<2s, No errors

2	Peak Load: Simulate heavy traffic during flood warning	1000	300	System handles load, Response time<5s
3	Stress Test:	5000	60	System does not crash,recovers gracefully
	Simulate sudden spike in alerts /SMS requests			



Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.6 seconds	Pass	System responded quickly under normal load
2	Response Time (Max)	< 5 seconds	4.1seconds	Pass	Slight delay during 1000-user peak load
3	Throughput (Req/sec)	200	250 req/sec	Pass	API handled high request volume for alerts
4	Error Rate	< 1%	Pass	Pass	Few timeouts during stress test spike
5	CPU Utilization	< 80%	Pass	Pass	CPU stable even at 5000 virtual users
6	Memory Utilization	< 80%	Pass	Pass	No memory leaks observed during tests

Step 4: Observations & Analysis

The system handled normal, peak, and stress loads effectively . Avverage response time was 4.2s, which is within the target.

Throughput of 250 req/sec shows the /api/ send alert endpoint can support multiple users during a flood event.Error rate was only 0.4%, confirming system stability.

BOTTLENECKS IDENTIFIED:

1.Database Query Delay: Slight **response time was seen when 1000+ users accessed flood status at the same time.**

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output)

Avg

Response Time



Response Time

1.6s

Max

4.2s



Response Time

Rising Waters

50t



Time Period	Value (t)
Start	50
1	55
2	53
3	55
4	58
5	58
6	60
7	59
8	60
9	63
10	64
11	65
12	68
13	69
14	70
15	72